Initial Notes for Boundary CFT

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Abstract

This is my primary learning notes for Boundary Conformal Field Theory (BCFT). This note stands for my initial understanding of the subject. It contains probably many assumptions and ideas. However, topics are proven to be correct in some sense(for I learn them from several books and lecture notes). Thus, I hope this note could be a good starting point for my future study of BCFT.

Contents

1 Convention Centre

Here I will explicitly list out the conventions used in this note.

- **Infinitesimal Conformal Transformation** We use the convention in the yellow book:

$$z \to z' = z + \epsilon(z) \tag{1}$$
$$\bar{z} \to \bar{z}' = \bar{z} + \bar{\epsilon}(\bar{z}) \tag{2}$$

  which gives:

$$\delta_{\epsilon,\bar{\epsilon}}\phi(z,\bar{z}) = -\left[\epsilon(z)\partial_z + h\partial_z\epsilon(z) + \bar{\epsilon}(\bar{z})\partial_{\bar{z}} + \bar{h}\partial_{\bar{z}}\bar{\epsilon}(\bar{z})\right]\phi(z,\bar{z}) \tag{3}$$

  For the primary fields.

- **Holomorphic coordinate** This is the standard definition

- **Holomorphic E-M tensor** there are two commonly used convention, we always take the yellow book convention:

$$T(z) = -2\pi T_{zz} \quad \bar{T}(\bar{z}) = -2\pi T_{\bar{z}\bar{z}} \tag{4}$$

  Different convention never changes the definition of $T(z)$ to ensure the same form of OPEs and Ward Identities.

- **Conformal Generators** are defined as the following comutation relation and the consistent exp map:

$$-[Q,\phi] = \delta_{\epsilon,\bar{\epsilon}}\phi \tag{5}$$

- **Relation between E-M tensor and Hamiltonian**

  Hamiltonian can be written in the following convention:

$$H = \int \frac{1}{2\pi} T_{tt} dx \tag{6}$$

  [I'll prove its consistency later.]

2 General Picture of BCFT

In ordinary CFT, we study the theory defined on a manifold without boundary. However, in many physical situations, we may need to consider the theory defined on a manifold with boundary. For example, in string theory, we may consider open strings which have two endpoints, these endpoints can be viewed as boundaries of the worldsheet. In condensed matter physics, we may consider critical systems with edges or surfaces, which can be viewed as boundaries of the bulk system. Thus, we need Boundary CFT to describe these situations.

> **Axiom 1.** ***General Picture of BCFT***
> *The construction of BCFT is induced from a ordinary CFT. We consider given a CFT defined on a plane. We call it the* ***Parent CFT****. Then we chop off a setionand make a boundary. We consider what is the behaiour of the theory...*
> *Note that inducing a parent CFT to a BCFT is never unique. Different boundary conditions may lead to different theories.*

2.1 BCFT on Upper Half Plane

In General we can study BCFT on any kinds of Boundary, but here we focus on the simplest case, BCFT on Upper Half Plane (UHP) $\mathbb{H}$. Considering UHP is general because of the Riemann Mapping Theorem:

> **Theorem 1.** ***Riemann Mapping Theorem***
> *Any simply connected open subset of the complex plane which is not the entire plane is conformally equivalent (exist a biholomophic map between) to the open unit disk:*
> $$D = \{z \in \mathbb{C} : |z| < 1\} \tag{7}$$

And we can map the unit disk to UHP via a Mobius Transformation:

$$w = \frac{z - i}{z + i} \quad z \in \mathbb{H}, w \in \mathbb{D}. \tag{8}$$

2.2 Conformal Transformation on UHP

Then we analyze the conformal transformation on the UHP. Due to the existence of the boundary, not all holomorphic transformations are allowed, only those preserving the boundary are allowed. Thus, we have:

> **Definition 1.** ***Conformal Transformation on UHP***
> *The conformal transformation on UHP are the holomorphic transformations $\epsilon(z)$ satisfying:*
> $$\mathrm{Im}(\epsilon(z)) = 0 \quad \textit{for } z \in \mathbb{R} \tag{9}$$

> **Remark:**
> We also have to notice that now $\epsilon(z)$ has its image and domain restricted to UHP $\mathbb{H}$ only.

Thus as expected the conformal algebra of a boundary CFT should be smaller than the conformal algebra of a CFT without boundary.

We often use a analytical continuation trick to reformulate the conformal transformation on UHP. We define:

Definition 2. ***Analytical continued conformal transformation***
The analytical continued conformal transformation $\epsilon(z)$ on the whole complex plane is defined as:

$$\epsilon^H(z) = \begin{cases} \epsilon(z) & \textit{if } \mathrm{Im}(z) > 0 \\ \bar{\epsilon}(z) & \textit{if } \mathrm{Im}(z) < 0 \end{cases} \tag{10}$$

This is mathematically available because of the boundary condition $\mathrm{Im}(\epsilon(z)) = 0$ on the real axis.

3 BCFT in Path Integral Formalism

3.1 Conformal Boundary Conditions

3.1.1 General Boundary Conditions

In ordinary QFT, if we have a boundary of the theory we need to fix a boundary condition. In ordinary QFT we have a dynamical field ϕ with action $S[\phi]$.

Then what we can do is to fix a boundary condition for the dynamical field and we're done once and for all. We may take Dirichlet Boundary Condition (DBC) or Neumann Boundary Condition (NBC) or any other boundary condition we like:

$$\text{DBC: } \phi|_{\partial M} = \phi_0, \tag{11}$$

$$\text{NBC: } \partial_n \phi|_{\partial M} = 0, \tag{12}$$

3.1.2 Conformal Boundary Condition

In CFT, we only have the Primaries and Descendants. We can't tell what they should satisfy on the boundary for general discussion (However, if we can write a Lagrangian for CFT, we of course can write those fields with dynamical field and fix a boundary condition). Thus, we focus on what is generally consistent, one condition is that:

- **Energy can't flow across the boundary**

Mathematically, if we consider the UHP then it is written as:

> **Definition 3.** ***Conformal Boundary Condition (Gluing Condition)***
> *On the boundary of a CFT:*
>
> $$\int_{-\infty}^{\infty} dx T_{01}(x,0) = 0 \tag{13}$$
>
> *or we assume locally:*
>
> $$T_{01}(x,0) = 0 \tag{14}$$
>
> *or equivalently in the holomorphic and anti-holomorphic coordinates :*
>
> $$T(z) = \bar{T}(\bar{z}) \quad \text{on } z = \bar{z} \tag{15}$$

3.1.3 Boundary Conditions

The Gluing Condition is never enough to define a CFT with boundary. Apparently, we need more information of the behavior of the fields on the boundary.

For example, for a CFT with a lagrangian description, eg. the free boson CFT, we can have DBC or NBC for the dynamical field ϕ on the boundary. These two boundary conditions will lead to different correlation functions and different physical behavior of the theory. Thus, apart from the Gluing Condition, we need more data to define a BCFT. Thus, we need some more conditions:

> **Definition 4.** ***Boundary Conditions in BCFT***
> *We assume that consider a BCFT induced from a CFT, there exist a set of **boundary conditions** $\{\alpha\}$ which consistently define the behavior of the fields.*

3.2 Ward Identities with Boundary

We recall the conformal Ward identity **??**:

$$\delta_{\epsilon,\bar{\epsilon}}\langle X\rangle = -\frac{1}{2\pi i}\oint_C dz\ \epsilon(z)\langle T(z)X\rangle - \frac{1}{2\pi i}\oint_{\bar{C}} d\bar{z}\ \bar{\epsilon}(\bar{z})\langle \bar{T}(\bar{z})X\rangle \tag{16}$$

Remark:
This Identity hold for any theory with conformal symmetry. When considering th e boundary, the only thing we need to change is:

- z can only take values in UHP and $\bar{z} = z^*$ thus can only take values in LHP.
- Transformation $\epsilon(z)$ and $\bar{\epsilon}(\bar{z})$ must preserve the boundary. Thus, they can't be independent and we can't decouple the holomorphic and anti-holomorphic parts any more.

Before we use the decoupling trick to make the holomorphic and anti-holomorphic parts independent. Now, we can't do this trick because of the existence of boundary. Thus, we have to deal with the two parts together.

3.2.1 Analytical Continuation Trick

We focus on the E-M tensors $T(z)$ and $\bar{T}(\bar{z})$ and $z \in \text{UHP}, \bar{z} = z^*$. In fact we now don't need two independent E-M tensors defined on half of the complex plane, we can define a single E-M tensor on the whole complex plane to encode the information of both $T(z)$ on the UHP and $\bar{T}(\bar{z})$ on LHP.

Definition 5. ***Analytical continued E-M Tensor***
we define the analytical continued E-M tensor $T(z)$ on the whole complex plane as:

$$T^H(z) = \begin{cases} T(z) & \textit{if}\ \mathrm{Im}(z) > 0 \\ \bar{T}(z) & \textit{if}\ \mathrm{Im}(z) < 0 \end{cases} \tag{17}$$

This is a holomorphic function on $\mathbb{C}$.

Remark:
Mathenatically, This is called the Analytic Continuation. And we can prove that this function is holomorphic on $\mathbb{C}$ mathematically.

Just as what we do for the E-M tensor in normal CFT, we can mode expand the analytical continued E-M tensor:

Definition 6. ***Mode Expansion of Analytical continued E-M Tensor***
The mode expansion of the analytical continued E-M tensor is defined as:

$$T^H(z) = \sum_{n\in\mathbb{Z}} z^{-n-2}L_n^H, \quad L_n^H = \frac{1}{2\pi i}\oint_C dz z^{n+1}T^H(z) \tag{18}$$

Remark:
Note that though we use the same notation of L_0 it is the mode expansion of the analytical continued E-M tensor now not just the holomorphic E-M tensor, but a combination of both holomorphic and anti-holomorphic E-M tensors.

3.2.2 Conformal Ward Identity on UHP

Thus, taking an appropriate contour shown in fig. 1 the RHS of the conformal Ward Identity can we written as following:

$$RHS = -\oint_{C^+} \frac{dz}{2\pi i}\epsilon(z)\langle T(z)X\rangle_H - \oint_{\bar{C}^+} \frac{d\bar{z}}{2\pi i}\bar{\epsilon}(\bar{z})\langle \bar{T}(\bar{z})X\rangle_H = -\oint_{C^+ + \bar{C}^+} \frac{dz}{2\pi i}\epsilon^H(z)\langle T^H(z)X\rangle_H \tag{19}$$

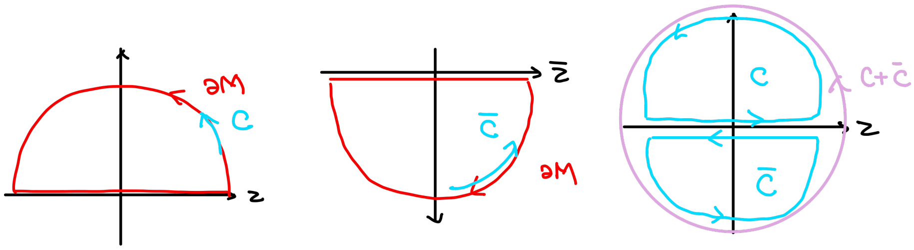

Figure 1: Choice of contour for conformal Ward identity on UHP after analytic continuation.

For Primary Field we can also do a similar trick to the LHS of the conformal Ward identity. We have:

$$\begin{aligned} LHS = &-\frac{1}{2\pi i}\oint_{C^+} dw\ \epsilon(w)\sum_{i=1}^{n}\left[\frac{1}{w-z_i}\partial_{z_i} + \frac{h_i}{(w-z_i)^2}\right]\langle X\rangle_H &(20)\\ &-\frac{1}{2\pi i}\oint_{\bar{C}^+} d\bar{w}\ \bar{\epsilon}(\bar{w})\sum_{i=1}^{n}\left[\frac{1}{\bar{w}-\bar{z}_i}\partial_{\bar{z}_i} + \frac{\bar{h}_i}{(\bar{w}-\bar{z}_i)^2}\right]\langle X\rangle_H &(21)\\ = &-\frac{1}{2\pi i}\oint_{C^+ + \bar{C}^+} dw\ \epsilon^H(w)\sum_{i=1}^{n}\left[\frac{1}{w-z_i}\partial_{z_i} + \frac{h_i}{(w-z_i)^2} + \frac{1}{w-\bar{z}_i}\partial_{\bar{z}_i} + \frac{\bar{h}_i}{(w-\bar{z}_i)^2}\right]\langle X\rangle_H &(22) \end{aligned}$$

Thus, we now have the conformal Ward identity on UHP:

Theorem 2. ***Conformal Ward Identity of Primaries on UHP***
On UHP, for a set of primary fields $X = \prod_{i=1}^{n}\phi_i(z_i,\bar{z}_i)$, we have the conformal Ward identity:

$$\langle T^H(z)X\rangle_H = \sum_{i=1}^{n}\left[\frac{1}{z-z_i}\partial_{z_i} + \frac{h_i}{(z-z_i)^2} + \frac{1}{z-\bar{z}_i}\partial_{\bar{z}_i} + \frac{\bar{h}_i}{(z-\bar{z}_i)^2}\right]\langle X\rangle_H \tag{23}$$

where $z \in \mathbb{C}$ and $T(z)$ is the analytical continued E-M tensor.

Remark:
Attention that on the above equation $z_i \in$ UHP however, $z \in \mathbb{C}$. they have different domains.

Additionally, we can also assume the conformal Ward identity holds for more general fields beyond primaries on UHP as well:

Axiom 2. ***Conformal Ward Identity on UHP***

On UHP, for a general set of fields X, we have the conformal Ward identity:

$$\delta_{\epsilon^H}\langle X\rangle_H = -\frac{1}{2\pi i}\oint_{C^+ + \bar{C}^+} \epsilon^H(z)\langle T^H(z)X\rangle_H \tag{24}$$

where $z \in \mathbb{C}$ and $T^H(z)$ is the analytical continued E-M tensor.

3.3 The Method of Images (Doubling Trick)

The above discussion have give us a hint that the coorelation on the UHP bahave as if we have twice the number of fields on the whole complex plane. We observe the difference between the whole conformal Ward identity on UHP and the holomorphic conformal ward identity on the whole complex plane for primary fields:

$$\langle T(z)X_0\rangle_{\mathbb{C}} = \sum_{i=1}^{n}\left[\frac{1}{z-z_i}\partial_{z_i} + \frac{h_i}{(z-z_i)^2} + \frac{1}{z-\bar{z}_i}\partial_{\bar{z}_i} + \frac{\bar{h}_i}{(z-\bar{z}_i)^2}\right]\langle X_0\rangle_{\mathbb{C}} \tag{25}$$

$$\langle T^H(z)X\rangle_H = \sum_{i=1}^{n}\left[\frac{1}{z-z_i}\partial_{z_i} + \frac{h_i}{(z-z_i)^2} + \frac{1}{z-\bar{z}_i}\partial_{\bar{z}_i} + \frac{\bar{h}_i}{(z-\bar{z}_i)^2}\right]\langle X\rangle_H \tag{26}$$

Where we have:

$$X_0 = \prod_{i=1}^{n}\phi_i(z_i)\prod_{i=1}^{n}\bar{\phi}_i(\bar{z}_i), \quad z_i, \bar{z}_i \in \mathbb{C}, z_i^* = \bar{z}_i \tag{27}$$

$$X = \prod_{i=1}^{n}\phi_i(z_i,\bar{z}_i), \quad z_i \in \text{UHP}, \bar{z}_i = z_i^* \in \text{LHP} \tag{28}$$

Where $\phi_i(z_i)$ are chiral fields with conformal dimension $(h_i, 0)$ and $\bar{\phi}_i(\bar{z}_i)$ are anti-chiral fields with conformal dimension $(\bar{h}_i, 0)$. We assume that they are all holomorphic fields.

- We can see that the **correlation function of the primaries on the UHP** satisfies the same differential equations as the **correlation function of twice of the number of chiral holomorphic fields on the whole complex plane**.

Remark:

By the word "chiral holomorphic fields" we mean the fields with conformal dimension $(h, 0)$ only and depends only on z.

In fact, chiral holomorphic fields can be physical, for example the majorana fermion field $\phi(z)$ can be a $(1/2, 0)$ chiral holomorphic field.

By the procudure of getting the 2 point and 3 point function, we know that the correlation function are determined by solving the conformal ward identities. Thus, we can conclude that:

Theorem 3. ***Method of Images (Doubling Trick)***

*The **correlation function** of a set of primary fields $X = \prod_{i=1}^{n}\phi_i(z_i,\bar{z}_i)$ on UHP*

$$=$$

$\sum$ *of the **chiral conformal blocks** of twice the number of chiral holomorphic fields $X_0 = \prod_{i=1}^{n}\phi_i(z_i)\prod_{i=1}^{n}\bar{\phi}_i(\bar{z}_i)$ on the whole complex plane.*

Proof: this is a straightforward result of the definition of conformal blocks and the above discussion.

An example is that the 1-point correlation function on UHP can be computed via the 2-point chiral conformal blocks on the whole complex plane:

$$\langle\phi(z,\bar{z})\rangle_H = \langle\phi(z)\bar{\phi}(\bar{z})\rangle_{\mathbb{C}} \sim \frac{1}{(z-\bar{z})^{2h}} \text{ for } h=\bar{h} \tag{29}$$

There will be some constant prefactor to be determined by the theory and the boundary condition on the boundary.

4 BCFT in Canonical Formalism

4.1 Radial Quantization with Boundary

In the context of BCFT, we can also perform the Radial Quantization. However, the equal-time slices are now semi-circles instead of full circles due to the presence of the boundary. Other tahn this the strategy of canonical quantization is similar to the ordinary CFT:

- Radial Ordering
- OPE and Commutators relation ...

4.1.1 Symmetry Algebra with Boundary

Then we start to study the symmetry algebra with boundary. We notice that from the general conformal Ward identity on UHP axiom 2, and the relation between the commutators and the OPE, we then have:

$$\delta_{\epsilon^H}\phi(w,\bar{w}) = -[Q_{\epsilon^H}, \phi(w,\bar{w})] = -\oint_C \frac{dz}{2\pi i}\epsilon^H(z)T^H(z)\phi(w,\bar{w}) \tag{30}$$

The integral contour is around the origin with both w and $\bar{w}$ inside of it. Then we can expand $\epsilon^H(z)$ and just as the ordinary CFT case we can give out the generators as:

$$L_n^H = \frac{1}{2\pi i}\oint_C dz\ z^{n+1}T^H(z) = \oint_{C^+}\frac{dz}{2\pi i}z^{n+1}T(z) + \oint_{\bar{C}^+}\frac{d\bar{z}}{2\pi i}\bar{z}^{n+1}\bar{T}(\bar{z}) \tag{31}$$

Just as the ordinary CFT case, we can compute the commutators between these generators via the OPE of $T^H(z)$ with itself. We have:

Theorem 4. ***Symmetry Algebra with Boundary***
The generators L_n^H defined above satisfy the Virasoro Algebra:

$$[L_n^H, L_m^H] = (n-m)L_{n+m}^H + \frac{c}{12}n(n^2-1)\delta_{n+m,0} \tag{32}$$

We can see that now the symmetry algebra of the BCFT is just a single copy of the Virasoro Algebra instead of two copies as in ordinary CFT.

4.1.2 Hamiltonian on a semi-circle

We then consider the Hamiltonian on a semi-circle. We consider the integral on a semi-circle. We name the contour as S which is the semi-circle part of C^+ with the same orientation as C^+ and the contour $\bar{S}$ which is the semi-circle part of $\bar{C}^+$ with the opposite orientation as $\bar{C}^+$.

The Hamiltonian of a theory should be given by the generator of the "time" diraction. In CFT radial quantization, it is defined to be the generator of the dilation transformation.

Remark:
It is important we define the Hamiltonian as the generator in time direction in the following style:

$$-\partial_t\psi = \delta\psi = -[H,\psi] \tag{33}$$

Stick to this convention will avoid many confusions.

Considering the BCFT case eq. (30), we have:

Theorem 5. ***Hamiltonian on Semi-circle***

The Hamiltonian on a semi-circle is defined as:

$$H^H = \frac{1}{2\pi i}\oint_S dz\ zT(z) + \frac{1}{2\pi i}\oint_{\bar{S}} d\bar{z}\ \bar{z}\bar{T}(\bar{z}) \tag{34}$$

$$= \frac{1}{2\pi i}\oint_{S+\bar{S}=C} dz\ zT^H(z) \tag{35}$$

$$= L_0^H \tag{36}$$

where L_0^H is the mode expansion of the analytical continued E-M tensor.

Proof: given by straightforward calculation, plug in the diation $\epsilon^H(z)$we can get this generator from eq. (30).

We will take this as a Hamiltonian of the BCFT.

4.1.3 Hilbert Space Structure with Boundary

Similarly, the Hilbert Space structure should take as the representation of a single copy of the Virasoro Algebra. Thus, we have:

Theorem 6. ***Hilbert Space Structure with Boundary***

The Hilbert Space of the BCFT is spanned by the highest weight representations of a single copy of the Virasoro Algebra:

$$\mathcal{H}_\alpha = \bigoplus_h n_\alpha^h \mathcal{V}_h \tag{37}$$

where n_h are non-negative integers indicating the multiplicity of the representation $\mathcal{V}_h$ in the Hilbert Space. Here we use α to indicate the boundary condition of the BCFT.

Then we expect to construct this Hilbert Space via acting operators on the vaccum state. However, we might face contraditions here, if we nively act a bulk primary field $\phi(z, \bar{z})$ on the vacuum state $|0\rangle$, then it transforms as a representation of two copies of the Virasoro Algebra instead of a single copy. Thus, the state-operator correspondence seems to be less obvious here.

4.2 Boundary Operators

4.2.1 Boundary Operators

In section section 3.3 we have seen that the one point correlation function on UHP can be computed via the two-point chiral conformal blocks on the whole complex plane. The correlation function of single bulk field behave divergence as it approaches the boundary. This phenomenon indicates that there may exist some new fields living on the boundary to interact with the bulk fields.

Remark:
In QFT, we understand divergence by revealing hidden existence of new fields.

Definition 7. ***Boundary Fields***

The fields living on the boundary of a BCFT are called Boundary Fields. We denote them as $\psi(x)$ where $x \in \mathbb{R}$ is the coordinate on the boundary. Here we have some defining conditions for the boundary fields:

- *They should be Chiral Fields with conformal dimension h only and can be decomposed into primaries and descendants.*
- *h should be in the parent CFT's operator algebra content*
- *OPE of primary boundary fields with the E-M tensor behave as:*

$$T^H(z)\psi(x) = \frac{h}{(z-x)^2}\psi(x) + \frac{1}{z-x}\partial_x\psi(x) + \textit{regular terms} \tag{38}$$

  we can only consider the holomorphic part approaching from the UHP.
- *OPE ensures that the commutation relation with the symmetry algebra as:*

$$[L_n^H, \psi(x)] = x^n(x\partial_x + (n+1)h)\psi(x) \tag{39}$$

4.2.2 Bulk-Boundary OPE

The divergence behavior of bulk fields approaching the boundary indicates that there should exist some relation between bulk fields and boundary fields. This relation is given by the Bulk-Boundary OPE:

Theorem 7. ***Bulk-Boundary OPE***
Bulk field should have the following behaviour as apporaching the boundary:

$$\phi(z,\bar{z}) \sim \sum_i C^{\alpha}_{(h,\bar{h})i}(2y)^{h_i-h-\bar{h}}\psi_i(x) \quad \textit{as } y \to 0 \tag{40}$$

where α indicates the boundary condition, $z = x+iy$, $\psi_i(x)$ are boundary fields with conformal dimension h_i and $C^{\alpha}_{\phi i}$ are the bulk-boundary structure constants depending on the boundary condition.

This is really a general form of a standar OPE of two fields. It is much an assumption that a derived result.

4.3 Hilbert Space from Boundary Fields

Now we can see that the construction of the Hilbert Space and the manifestation of the state-operator correspondence in BCFT should be done via the boundary fields instead of the bulk fields. We have:

Theorem 8. ***State Operator Correspondence in BCFT***
The Hilbert Space of the BCFT can be constructed via acting boundary fields on the vacuum state:

$$|h\rangle = \psi_h(0)\,|0\rangle \tag{41}$$

This construction is valid for:

- The state $|h\rangle$ indeed is a primary state with conformal dimension h. due to the commutation relation between the boundary fields and the symmetry algebra.

- The Bulk Operator can be expanded via the boundary operators as it approaches the boundary due to the Bulk-Boundary OPE.

Thus we believe that the Hilbert Space structure given above is valid.

4.4 CFT on a Strip

4.4.1 Transform to a Strip

Now consider a more general case of boundary condition. We never said that the boundary can only contain 1 single boundary condition. We then have to think about what happens if there are more than 1. To motivate the picture, we conformally map the CFT on the UHP minus the origin to a strip. We define the following mapping.

Definition 8. ***Mapping to a Strip***
We define the conformal transformation of $\mathbb{H}/\{0\} \to$ *Strip as the following:*

$$t - i\sigma = w = \frac{L}{\pi} ln(z), \quad x + iy = z \in \mathbb{H} \tag{42}$$

The following diagram illustrate the process:

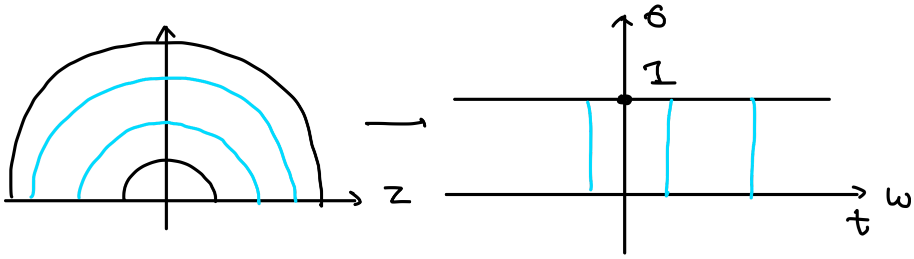

Figure 2: Transformation to a Strip (there's a mistake in the diagram, it should be $-\pi \to 0$ on the σ axis)

We then transform the fields on the UHP to the Strip:

$$\phi^S(w,\bar{w}) = \left(\frac{dw}{dz}\right)^{-h}\left(\frac{d\bar{w}}{d\bar{z}}\right)^{-\bar{h}} \phi^H(z,\bar{z}) = (\alpha z)^h (\alpha \bar{z})^{\bar{h}} \phi^H(z,\bar{z}), \qquad \alpha = \frac{\pi}{L}, \tag{43}$$

$$T^S(w) = \left(\frac{dw}{dz}\right)^{-2}\left(T(z) - \frac{c}{12}\{w,z\}\right) = \alpha^2\left(z^2 T(z) - \frac{c}{24}\right), \tag{44}$$

$$\bar{T}^S(\bar{w}) = \left(\frac{d\bar{w}}{d\bar{z}}\right)^{-2}\left(\bar{T}(\bar{z}) - \frac{c}{12}\{\bar{w},\bar{z}\}\right) = \alpha^2\left(\bar{z}^2 \bar{T}(\bar{z}) - \frac{c}{24}\right). \tag{45}$$

Note that here we use the concrete $T(z)$ and $\bar{T}(\bar{z})$ instead of the analytical continued one for clarity.

4.4.2 Hamiltonian on a Strip

Remark:
What is the Hamiltonian?
There are many seemingly equivalent definitions of the Hamiltonian in a QFT:

- Legendre Transformation of the Lagrangian

- Generator of the time translation
- Integral of the energy density T_{tt}

The first one and the third one in trivially equivalent with the definition. However, naively, the legrendre transformation may not lead to the generator of time translation, with a lack of coefficient.

However, in QM the importance of Hamiltonian is from the definition as the generator of time translation. Thus, we should always define the Hamiltonian as the generator of time translation as a fundamental definition and fix the convention from this!!

Then, we can compute the Hamiltonian on the strip, we have:

Theorem 9. *Hamiltonian on a Strip*

The Hamiltonian on a strip with width π is given by:

$$H^S_{\alpha\beta} = \frac{\pi}{L}\left(L_0^H - \frac{c}{24}\right) \tag{46}$$

where L_0^H is the mode expansion of the analytical continued E-M tensor and α, β indicate the boundary conditions on the two boundaries of the strip.

Proof: The Hamiltonian on the strip is given by the generator of the translation along t direction. We have:

$$H^S_{\alpha\beta} = \int_0^{-L} d\sigma\left(T_{tt}(\sigma, t)\right), \quad T_{tt} = T_{zz} + T_{\bar{z}\bar{z}} = \frac{1}{-2\pi}(T(w) + \bar{T}(\bar{w})) \tag{47}$$

[I come up with this argument myself, please have someone to check this with me!!!] Thus we have:

$$\begin{aligned} H^S_{\alpha\beta} &= \int_0^{-L} \frac{1}{2\pi i} d\omega T^S(w) + \int_0^{-L} \frac{1}{2\pi i} d\bar{\omega}\bar{T}^S(\bar{w}) && (48)\\ &= \frac{1}{2\pi i}\frac{\pi}{L}\int_{C+} dzzT(z) + \frac{1}{2\pi i}\frac{\pi}{L}\int_{\bar{C}+} d\bar{z}\bar{z}\bar{T}(\bar{z}) - \frac{1}{2\pi}\frac{c}{12}\int_0^L \frac{\pi}{L} d\sigma && (49)\\ &= \frac{\pi}{L}\left(L_0^H - \frac{c}{24}\right) && (50)\end{aligned}$$

Notice that here we use $dw = \dfrac{L}{\pi}\dfrac{dz}{z}$

Remark:
What is the Hamiltonian??

The hamiltonian is the generator of translation along the time direction. so we should define it as some operators that satisfy:

$$[H, \phi(\sigma, t)] = \partial_t \phi(\sigma, t) \tag{51}$$

this is a convention in CFT.

Remark:

Sometimes we define the conformal transformation to a strip with width $L = \pi$. In this case the Hamiltonian will be:

$$w = ln(z) \quad \Rightarrow \quad H^S_{\alpha\beta} = L_0^H - \frac{c}{24} \tag{52}$$

This is also a common convention. In the rest of this note we will use this convention for simplicity.

Remark:
We note that L_n operator only takes the interpretation of conformal generator **On the Plane Coordinate**. Though we can use these operator to represent the Hamiltonian on the strip.

4.5 Boundary Changing Operators

4.5.1 Boundary Changing Operators

Then it motivates to think that what happens if we have different boundary conditions on the two boundaries of the strip. In this case, if we map the theory back to the UHP, we will have 2 boundary conditions on the real axis with a shift in the origin. To capture the divergence on the origin, we need to introduce a new type of operators called Boundary Changing Operators.

Definition 9. ***Boundary Changing Operators***
The operators that change the boundary condition from α to β at a point on the boundary are called Boundary Changing Operators. We denote them as $\psi_{\alpha\beta}(x)$ where $x \in \mathbb{R}$ is the coordinate on the boundary.

[To be continued ... I don't fully understand this topic yet ...]

4.5.2 Hilbert Space with 2 Boundaries

The Hilbert Space with 2 boundaries may primarily be different, we use the following general notation to stand for it:

Definition 10. ***Hilbert Space with 2 Boundaries***
The Hilbert Space of the BCFT with boundary conditions α and β on the two boundaries is denoted as $\mathcal{H}_{\alpha\beta}$:

$$\mathcal{H}_{\alpha\beta} = \bigoplus_h n^h_{\alpha\beta}\mathcal{V}_h \tag{53}$$

Here we use the notation $n_h^{\alpha\beta}$ to indicate the multiplicity of the representation $\mathcal{V}_h$ in the Hilbert Space.

An explanation of why this might fall into Virasoro representation is that we always assume that boundary condition is compatible with the conformal symmetry.

However, having two boundary conditions makes the translation symmetry along the boundary break down, thus we might not be that sure about this.

4.5.3 State-Operator Correspondence

In this case, the conformal symmetry seems to break down more, we don't have translation symmetry along the boundary any more.

- In this case, there won't be a well-defined state-operator correspondence any more because of the breaking of translation symmetry along the boundary.

5 Boundary Conditions of BCFT

[I think Cardy's discussion in this topic is confusing, I'd like to re-derive it in my own way ... Please have someone to check this with me!!!!!!]

In above sections we learn that there are additional objects in BCFT compared to the parent CFT, such as boundary conditions, boundary fields, boundary changing operators. However, these objects can not arbitrarily exist, they should satisfy some consistency conditions to make the theory well-defined.

This section we will discuss what kinds of consistent boundary conditions can exist for a given parent CFT. To charaterize the boundary condition, we have a very elegant way of describing it, which is asigning a special state to a boundary condition. This special state is called the Boundary State.

5.1 Partition Function of BCFT on a Strip

5.1.1 Partition Function on a Strip

To construct a state that encode the boundary condition, we have to consider a thermal partition function of the CFT. It is much more convenient to define it with the cooridnate transformation to a strip. Thus, we define the following partition function:

Definition 11. ***Partition function of CFT on a Strip***

Consider a CFT on a UHP with conformal generators $\{L_n^H\}$. The thermal partition function on a strip with boundary conditions α and β on the two boundaries is defined as:

$$Z_{\alpha\beta}(\tau) = \mathrm{Tr}_{\mathcal{H}_{\alpha\beta}} e^{-2\pi(L_0^H - c/24) Im\tau} = \mathrm{Tr}_{\mathcal{H}_{\alpha\beta}} q^{L_0^H - c/24} \tag{54}$$

where τ is the modular parameter of the strip and $q = e^{2\pi i\tau}$.Here we have τ to be pure imaginary. The geometry of the compactified strip defined above is one with circumference $2\pi Im(\tau)$ and length π.

Definition 12. ***Partition Function of CFT on a Strip (General Definition)***

In General, we can define a thermal partition function on a strip with the following statement. Consider a theory with Hamiltonian $H^S_{\alpha\beta}$ on a strip with boundary conditions α and β on the two boundaries, which means that:

$$[H^S_{\alpha\beta}, \psi(x)] = \partial_t \psi(x) \tag{55}$$

We define the thermal partition function on the strip as:

$$Z_{\alpha\beta}(\tau) = \mathrm{Tr}_{\mathcal{H}_{\alpha\beta}} e^{-\tau H^S_{\alpha\beta}} \tag{56}$$

Where, τ characterizes the circumference of the compactified direction.

This thermal partition function has a geometric interpretation. It can be viewed as a partition function on a cylinder where the compartified direction is the Euclidean time direction and the spatial direction is the strip direction. The two boundaries of the strip are then the two boundaries of the cylinder with boundary conditions α and β respectively.

5.1.2 Characters of Virasoro Algebra

Then we can expand the partition function via the characters of the representations of the Virasoro Algebra:

Definition 13. ***Character of Virasoro Algebra***

The character of a representation $\mathcal{V}_h$ of the Virasoro Algebra is defined as:

$$\chi_h(q) = \mathrm{Tr}_{\mathcal{V}_h} q^{L_0 - c/24} \tag{57}$$

$$= q^{h-c/24} \sum_{N=0}^{\infty} p(N) q^N \tag{58}$$

where $p(N)$ is the number of partitions of integer N.

[please check this definition ... I'm not sure whether its correct ...]

There is a Important property of the characters under modular transformation:

Theorem 10. ***Modular Transformation of Characters***

Under the modular transformation $\tau \to -1/\tau$, the characters transform as:

$$\chi_h(e^{-2\pi i/\tau}) = \sum_{h'} S_h^{h'} \chi_{h'}(e^{2\pi i\tau}) \tag{59}$$

where $S_h^{h'}$ is the modular S-matrix of the parent CFT. Note that χ_h is the same function.

It just shows that the characters form a representation of the modular group, it behaves as a Covariant object under modular transformation.

We then can see that the thermal partition function can be expanded via the characters:

$$Z_{\alpha\beta}(q) = \sum_h n^h_{\alpha\beta} \chi_h(q) \tag{60}$$

Where $n^h_{\alpha\beta}$ are non-negative integers indicating the multiplicity of the representation $\mathcal{V}_h$ in the Hilbert Space $\mathcal{H}_{\alpha\beta}$.

5.2 CFT on a (choped) Cylinder

5.2.1 Transformation to a Cylinder

Apart from the above picture of thermal BCFT on a strip, now I want to define a **Choped Parent CFT on a Annulus**. Consider the **Parent CFT** which generator this BCFT. We can transform it to a cylinder and chop it to match the geometry of the compactified strip. We then have to define a parent CFT on a cylinder:

Definition 14. ***Mapping Parent CFT to a Cylinder***

The mapping from the complex plane to a cylinder is defined as:

$$t - i\sigma = \xi = \frac{L}{2\pi} \ln z, \quad z = x + iy \in \mathbb{C} \tag{61}$$

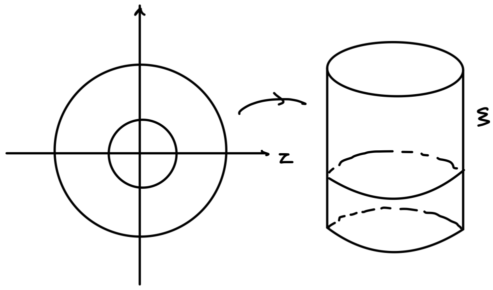

Figure 3: Mapping Parent CFT to a Cylinder

Under this mapping the E-M tensor transforms as:

$$\phi^C(\xi,\bar{\xi}) = \left(\frac{d\xi}{dz}\right)^{-h}\left(\frac{d\bar{\xi}}{d\bar{z}}\right)^{-\bar{h}}\phi^P(z,\bar{z}) = \left(\frac{2\pi}{L}\right)^{h+\bar{h}} z^h\bar{z}^{\bar{h}}\phi^P(z,\bar{z}) \tag{62}$$

$$T^C(\xi) = \left(\frac{d\xi}{dz}\right)^2\left[T^P(z) - \frac{c}{12}\{\xi,z\}\right] = \left(\frac{2\pi}{L}\right)^2\left[z^2T^P(z) - \frac{c}{24}\right] \tag{63}$$

$$\bar{T}^C(\bar{\xi}) = \left(\frac{d\bar{\xi}}{d\bar{z}}\right)^2\left[\bar{T}^P(\bar{z}) - \frac{c}{12}\{\bar{\xi},\bar{z}\}\right] = \left(\frac{2\pi}{L}\right)^2\left[\bar{z}^2\bar{T}^P(\bar{z}) - \frac{c}{24}\right] \tag{64}$$

Remark:
Recal different E-M tensor used in this section:
E-M tensor of Parent CFT on complex plane: $T^P(z)$;
E-M tensor of Parent CFT on cylinder: $T^C(\xi)$;
E-M tensor of BCFT on UHP: $T^H(z)$;
E-M tensor of BCFT on strip: $T^S(w)$.

5.2.2 Hamiltonian of Parent CFT on a Cylinder

Then we derive the Hamiltonian on the cylinder:

Theorem 11. ***Hamiltonian of Parent CFT on Cylinder***
The Hamiltonian of the Parent CFT on a cylinder with circumference L is given by:

$$H^C = \frac{2\pi}{L}\left(L_0^P + \bar{L}_0^P - \frac{c}{12}\right) \tag{65}$$

where $L_0^P, \bar{L}_0^P$ are the mode expansions of the holomorphic and anti-holomorphic E-M tensors of the Parent CFT on a complex plane.

Proof: The Hamiltonian on the cylinder is given by the generator of the translation along t direction. We have:

$$H^C = \int_0^{-L} d\sigma\,(T_{tt}(\sigma,t)) \tag{66}$$

Then substituting the relation between T_{tt} and $T^C(\xi), \bar{T}^C(\bar{\xi})$ we have:

$$T_{tt} = T_{\xi\xi} + T_{\bar{\xi}\bar{\xi}} = \frac{1}{-2\pi}(T^C(\xi) + \bar{T}^C(\bar{\xi})) \tag{67}$$

Thus we have:

$$H^C = \int_0^{-L} \frac{1}{2\pi i} d\xi T^C(\xi) + \int_0^{-L} \frac{1}{2\pi i} d\bar{\xi}\bar{T}^C(\bar{\xi}) \tag{68}$$

$$= \frac{1}{2\pi i}\frac{2\pi}{L}\int_C dz z T^P(z) + \frac{1}{2\pi i}\frac{2\pi}{L}\int_{\bar{C}} d\bar{z}\bar{z}\bar{T}^P(\bar{z}) - \frac{1}{2\pi}\frac{c}{12}\int_0^L \frac{2\pi}{L} d\sigma \tag{69}$$

$$= \frac{2\pi}{L}\left(L_0^P + \bar{L}_0^P - \frac{c}{12}\right) \tag{70}$$

[I come up with this argument myself, please have someone to check this with me!!!]

> **Remark:**
> **Discussion on the Normalization of the Hamiltonian on a Cylinder**
>
> The above proof contains many gaps, and in fact no standard textbook provides a fully rigorous derivation. The essential subtlety lies in the definition and normalization of the Hamiltonian on the cylinder. In practice, the Hamiltonian is defined with a specific normalization factor, which is finally fixed by demanding that the commutation relations generate the correct geometric transformations.
>
> We know that a correct generator must produce the correct transformation of fields through its commutators. Here, the Hamiltonian should generate translations along the Euclidean time direction t on the cylinder. Thus we require
>
> $$[H^C_{\alpha\beta}, \phi^C(\xi,\bar{\xi})] = \partial_t \phi^C(\xi,\bar{\xi}) = (\partial_\xi + \partial_{\bar{\xi}})\phi^C(\xi,\bar{\xi}). \tag{71}$$
>
> This requirement fixes the normalization of the Hamiltonian. One may then verify that the definition of $H^C_{\alpha\beta}$ in terms of Virasoro generators indeed satisfies this condition. Therefore it is a valid Hamiltonian on the cylinder.
>
> However, checking this explicitly requires computing the commutator between L_0 and the cylinder field $\phi^C(\xi,\bar{\xi})$. This step is subtle because we must use
>
> $$[L_0, \phi^C(\xi,\bar{\xi})] = [L_0^H, (\alpha z)^h(\alpha\bar{z})^{\bar{h}}\phi^P(z,\bar{z})] \tag{72}$$
>
> $$= (\alpha)^{h+\bar{h}} z^h \bar{z}^{\bar{h}} (z\partial_z + h)\phi^P(z,\bar{z}), \tag{73}$$
>
> where $\alpha = \frac{2\pi}{L}$ and $z = e^{\alpha\xi}$.
>
> Now observe the identity
>
> $$z\partial_z\big(z^h\phi^P(z,\bar{z})\big) = hz^h\phi^P(z,\bar{z}) + z^h(z\partial_z\phi^P(z,\bar{z})),$$
>
> which implies
>
> $$[L_0, \phi^C(\xi,\bar{\xi})] = z\partial_z\big((\alpha)^{h+\bar{h}} z^h\bar{z}^{\bar{h}}\phi^P(z,\bar{z})\big) = \partial_\xi\phi^C(\xi,\bar{\xi}). \tag{74}$$
>
> Thus the Hamiltonian must be proportional to the first-order operators:
>
> $$H = L_0 + \bar{L}_0 + \text{constant}$$
>
> and this fixes the overall normalization!!! Then we say that the overall constant comes from the transformation of the E-M tensor.

> **Remark:**
> **What is the Hamiltonian??**
>
> The hamiltonian is the generator of translation along the time direction. so we should define it as some operators that satisfy:
>
> $$[H, \phi(\sigma,t)] = \partial_t\phi(\sigma,t) \tag{75}$$

∎ this is a convention in CFT.

[This important proof can be better organized]

5.2.3 Partition Function of a boundary Parent CFT on a Cylinder

Then we can define the partition function of the Parent CFT on a cylinder with states assigned to two boundaries:

Definition 15. ***Partition function of Parent CFT on a Choped Cylinder***
Consider a Parent CFT on a complex plane with conformal generators $\{L_n^P\}, \{\bar{L}_n^P\}$. The partition function on a cylinder with states $|\alpha\rangle, |\beta\rangle$ assigned to the two boundaries and the geometry of length a and circumfence L is defined as:

$$Z_{\alpha\beta}(a, L) = \langle\alpha|e^{-aH^C}|\beta\rangle \quad \textit{where } H^C = \frac{2\pi}{L}\left(L_0^P + \bar{L}_0^P - \frac{c}{12}\right) \tag{76}$$

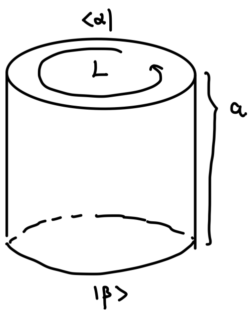

Figure 4: Partition Function on a Cylinder

However, this is a theory with boundaries, thus we need to impose the gluing condition on the boundaries for the E-M tensor. There are two different derivations of this condition:

- Transform it into a UHP and transform the boundary to the real axis. Then impose the condition that there is no energy flow across the boundary, which gives $T(z) = \bar{T}(\bar{z})$ on the boundary. Transforming it back to the cylinder gives the gluing condition.

Remark:

We shall note that we can't use the transformation $w = \ln(z)$ here ,we have to let the boundary to be the real axis.

- Consider that no energy flows across the boundary directly on the cylinder, which gives the same condition.

The condition is given by:

Theorem 12. ***Gluing Condition on a Cylinder***
The gluing condition on a cylinder with boundaries assigned states $|\alpha\rangle, |\beta\rangle$ is given by:

$$(L_n^P - \bar{L}_{-n}^P)|\alpha\rangle = 0, \quad (L_n^P - \bar{L}_{-n}^P)|\beta\rangle = 0 \tag{77}$$

THIS THEOREM IS VERY SUBTLE AND IMPORTANT, MANY LITERATURE DO NOT GIVE A COMPREHENSIVE PROOF, PLEASE CHECK THIS!!!

Proof: We want to proof the relation by transforming the gluing condition on the UHP to a annulus on a complex plane. where the boundary meets the boundary of the troped cylinder. just from a to c in the following diagram.

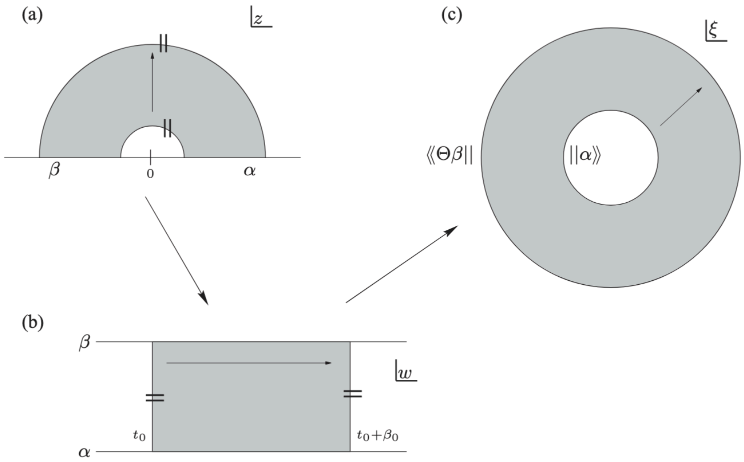

Figure 5: Transformation from Cylinder to UHP

- **Inner circle transfromation:** Inner circle $|\xi| = 1$ maps to the boundary at $z \in \mathbb{R}_+$ of the cylinder;

We define the transformation for the inner circle from as:

$$\xi = e^{\frac{2\pi i}{L}\ln z} \quad \bar{\xi} = e^{-\frac{2\pi i}{L}\ln \bar{z}} \tag{78}$$

This transformation has the following property on the boundary:

$$\bar{\xi} = \xi^{-1}, \quad \left(\frac{d\bar{\xi}}{d\bar{z}}\right)\left(\frac{d\xi}{dz}\right)^{-1} = -\bar{\xi}^2, \quad \{\xi, z\} = \{\bar{\xi}, \bar{z}\}, z = \bar{z} \tag{79}$$

Thus the gluing condition on the UHP $T^P(z) = \bar{T}^P(\bar{z})$ transforms to:

$$\left[\left(\frac{d\xi}{dz}\right)^2 T^P(\xi) - \left(\frac{d\bar{\xi}}{d\bar{z}}\right)^2 \overline{T}^P(\bar{\xi})\right] = 0 \tag{80}$$

Then we use the above properties and mode expand the gluing condition, we can see that:

$$T^P(\xi,\bar{\xi}) - \bar{\xi}^4\bar{T}^P(\bar{\xi},\xi) = \sum_n \left(L_n^P - \bar{L}_{-n}^P\right)\xi^{-n-2} = 0 \tag{81}$$

Finally, we can get that the gluing condition on the cylinder is given by:

$$(L_n^P - \bar{L}_{-n}^P)|\alpha\rangle = 0, \quad (L_n^P - \bar{L}_{-n}^P)|\beta\rangle = 0 \tag{82}$$

- **Outerier circle transformation:**

[I still don't understand this proof very well... but it is claim that this won't give out any new condition]

5.2.4 Ishibashi States

States that satisfy the gluing condition are called Ishibashi States. They can be constructed as follows:

Theorem 13. ***Ishibashi States***

For each representation $\mathcal{V}_h$ of the Virasoro Algebra of the Parent CFT, there exists a unique Ishibashi State $|h\rangle\rangle$ satisfying the gluing condition:

$$|h\rangle\rangle = \sum_N |h;N\rangle \otimes \overline{|h;N\rangle} \tag{83}$$

5.3 Boundary State Formalism

5.3.1 Definition of Boundary State

With the preparation above, we then can fully define the Boundary State. It is defined as follows:

Definition 16. ***Boundary State***

The boundar state is defined as a state on a boundary parent CFT on a cylinder $|\alpha\rangle, |\beta\rangle$

- *that has the same geometry as a compactified strip;*
- *and gives the same partition function as the thermal partition function of the BCFT on the strip with boundary condition α, β.*

Here we assume the cylinde has length π and circumfence $2\pi Im(\tau)$ for simplicity. Thus, the boundary state $|\alpha\rangle, |\beta\rangle$ assigned to the two boundaries is defined as:

$$Z_{\alpha\beta}(\tau) = \mathrm{Tr}_{\mathcal{H}_{\alpha\beta}} q^{L_0^H - c/24} = \langle\alpha|e^{-\pi H^C}|\beta\rangle \quad \textit{where } H^C = \frac{1}{Im(\tau)}\left(L_0^P + \bar{L}_0^P - \frac{c}{12}\right) \tag{84}$$

The right hand side can also be written in a $q = e^{2\pi i\tau}$ style:

$$Z_{\alpha\beta} = \langle\alpha|(\tilde{q}^{1/2})^{L_0^{(P)} + \overline{L}_0^{(P)} - c/12}|\beta\rangle, \tag{85}$$

where $\tilde{q} = e^{-2\pi i/\tau} = e^{-2\pi/Im(\tau)}$.

Boundary state is the state lives on the boundary of the Parent CFT on a cylinder that encodes the boundary condition of the BCFT on a strip. Thus, it should satisfy the gluing condition:

$$(L_n^P - \bar{L}_{-n}^P)|\alpha\rangle = 0, \quad (L_n^P - \bar{L}_{-n}^P)|\beta\rangle = 0 \tag{86}$$

So we have the conclusion that:

Theorem 14. ***Boundary State Expansion***

The boundary state can be expanded via the Ishibashi states as:

$$|\alpha\rangle = \sum_h |h\rangle\rangle\langle\langle h|\alpha\rangle \tag{87}$$

5.3.2 Cardy Condition

If the Parent CFT is a Diagnonal CFT, we can derive a very important condition for the boundary states called the Cardy Condition and give out an important solution to it. For a diagonal CFT, we have $h = \bar{h}$ for all primary fields.

Due to the property of the Diagonal CFT, we can rewrite the partition function on a troped cylinder as:

$$Z_{\alpha\beta} = \langle\alpha|(\tilde{q}^{1/2})^{L_0^{(P)}+\overline{L}_0^{(P)}-c/12}|\beta\rangle = \sum_h \langle\alpha|h\rangle\rangle\langle\langle h|\beta\rangle\chi_h(\tilde{q}) \tag{88}$$

Compare with the partition function on a strip and using the modular transformation of characters, we have:

Theorem 15. ***Cardy Condition***

The boundary states $|\alpha\rangle, |\beta\rangle$ must satisfy the following condition:

$$\sum_h \langle\alpha|h\rangle\rangle\langle\langle h|\beta\rangle S_h^{h'} = n_{\alpha\beta}^{h'} \tag{89}$$

or equivalently:

$$\langle a|h'\rangle\rangle\langle\langle h'|b\rangle = \sum_h S_h^{h'} n_{ab}^h. \tag{90}$$

6 Summary

I know, above notes are not rigorous. It's filled with arguments and assumptions that are doubtful. Now, I want to discuss how should we actually think of a BCFT.

From my perspective, its messy to think of it as a ordinary QFT with boundary, we analyze the boundary conditions and do canonical quantization, find spectrums and correlation functions, etc.

Instead, just as CFT do, 1. we initially use the traditional QFT style and tons of messy assumptions to find a possible structure or the theory, 2. and then we take these structures as axioms and try to build a consistent theory. The second step is beyond the scope of this note, but I think one should know there's this step to take. And if time allows, I'll continue this adventure in another note.